

Daniel Astudillo

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EXPERIENCE

Software Engineer II

S&P Global Inc.

October 2025 - Present, New York, NY

- Guided **Data as a Service (DaaS)** frontend modernization effort to **NextJS 16.1.5, React 19.2.4, and Tailwind v4**, **reducing build size by 70%** and **addressing 10+ critical-severity security vulnerabilities**.
- Leveraged **Google BigQuery Storage Write API v2** and **gRPC** to replace **BigQuery DML operations** for Data and **reduce average query execution time by ~60% from 21 secs to 10 secs**.
- Migrated **DaaS API** from **Google BigQuery Storage Write API v2 + BigQuery SQL hybrid** to **EF Core + PostgreSQL** to further **reduce average query execution time from 10 secs to <2 secs**.
- Achieved **96% code coverage** and resolved **100+ high and medium-severity security vulnerabilities** in **DaaS API and Gateway**.

Software Engineer

Visa Inc.

September 2022 - February 2025, Miami, FL

- Modernized the **Visa Card Eligibility Service (20M+ monthly requests)**, upgrading to **.NET 6 and Spring Boot 3**, **improving critical-path API latency by 18%** and **cutting average response time by 15%**.
- Rebuilt **200+ T-SQL stored procedures** into APIs, decreasing **95th-percentile latency by 40%** and **maintenance overhead by 60%**, streamlining backend operations.
- Developed the first real-time benefit redemption processing system between **VCES** and the **Visa Digital Benefits Platform (VDBP)**, handling **100K+ daily events** with **Dockerized ActiveMQ Artemis message broker instances**, a **.NET Core AMQPS producer (VCES)**, and a **Spring Boot Jakarta JMS consumer (VDBP)**, achieving **99.99% uptime** and **sub-200ms event latency**.
- Migrated **Visa Click to Pay** to **Next.js + React**, achieving **40% faster page loads** and **30% fewer API errors**, which enhanced checkout **reliability** and **scaled seamlessly for 10M+ global users**.
- Engineered a **PCI-compliant data masking service** protecting sensitive cardholder information **in transit and at rest** for **VCES** to support regulatory requirements and bolster platform security.

PROJECTS

Real-Time Collaborative Editor

Personal Project • 2025

- Developed a real-time collaborative editor with a **React + Vite** frontend and **Node.js** backend supporting multi-user simultaneous editing with **CRDT-based conflict-free synchronization**, improving **edit latency to under 80ms**.
- Designed a production-grade MVP with **JWT authentication** via **Clerk**, and managed frontend/backend deployments on **Vercel** and **Fly.io**, respectively, improving **scalability and reducing infrastructure setup time by 60%**.

GenAI Contract Assistant

Visa • 2024

- Built a full-stack GenAI chatbot (Django frontend, Flask backend) for automated contract redlining, reducing manual review time by 30%.
- Implemented a **retrieval-augmented generation (RAG) pipeline** with LangChain, achieving 99% retrieval accuracy.
- Advanced the prototype to a company-supported initiative, demonstrating enterprise GenAI adoption potential.

EDUCATION

B.A. in Computer Science & Mathematics

Williams College • Williamstown, MA • June 2022

SKILLS

Languages: Java, C#, Python, JavaScript (ES6+), TypeScript, T-SQL, HTML/CSS

Frameworks: Spring Boot, .NET Core, EF Core, Flask, Django, React, Next.js, Vite, Node.js, Socket.IO, Angular, LangChain, Pandas, NumPy

Databases: PostgreSQL, MSSQL, MongoDB, IBM DB2, Google BigQuery

Cloud & Tools: AWS (EC2, S3), Docker, NGINX, Git, Jenkins, GKE, Azure DevOps, k8s

Testing: JUnit, Mockito, Jest, unittest, pytest

GenAI/ML: OpenAI API, Retrieval-Augmented Generation (RAG), LLM Prompt Engineering, scikit-learn